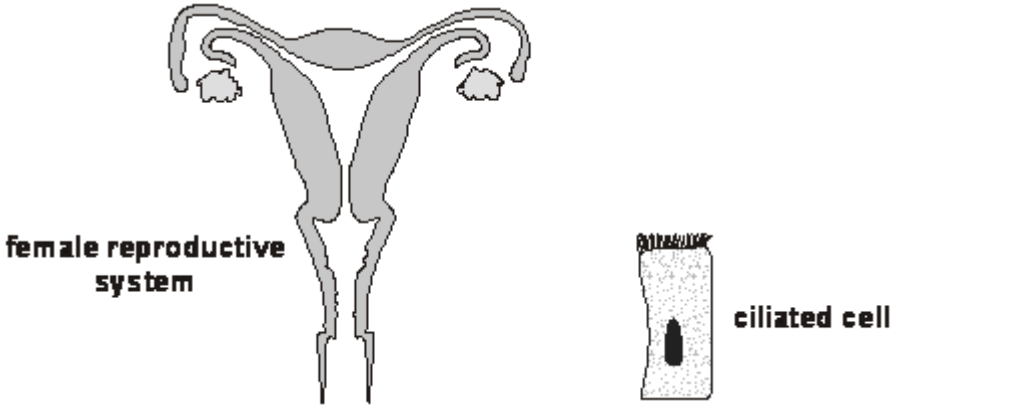
Q1.

(a) The diagram below shows the female reproductive system and a ciliated cell.



not to scale

Ciliated cells move an ovum along part of the reproductive system.

(i) In which part of the reproductive system are ciliated cells found?

.....

1 mark

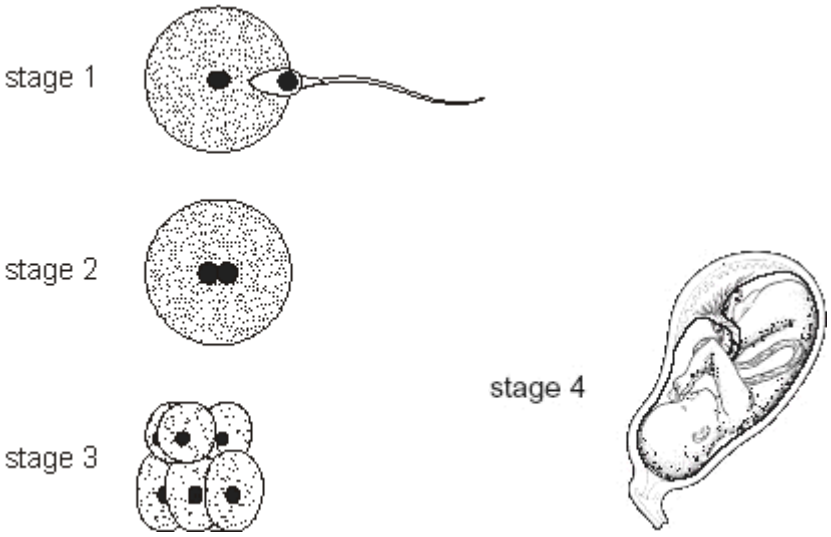
(ii) Describe how ciliated cells move an ovum along.

.....

.....

1 mark

(b) The diagrams below represent what happens at fertilisation and after fertilisation has taken place.



not to scale

(i) Some women find it difficult to become pregnant. Doctors have developed a technique in which an ovum is fertilised in a test-tube. An embryo is then implanted into the woman's reproductive system.

Which stage in part (b) shows an embryo and which stage shows a foetus?

embryo .....

foetus .....

1 mark

(ii) Into which part of the woman's reproductive system is the embryo implanted?

.....

1 mark

(c) (i) Explain why a child can look like both parents but is **not** identical to either of the parents.

.....

.....

.....

.....

2 marks

(ii) In the table below, tick **one** box by each human characteristic to show whether it is:

- inherited only
- inherited **and** affected by environmental conditions.

human characteristic	**inherited only**	**inherited and affected by environmental conditions**
eye colour		
skin colour		
weight		

1 mark

maximum 7 marks

Q1.

(a) (i) oviduct **or** fallopian tube

1 (L7)

(ii) cilia move in a wave **or** from side to side

*accept 'they waft **or** carry it along'*
accept 'the hairs move it along'
*accept 'the hairs **or** cilia waft **or** move'*
'the hairs' or 'the cilia' are insufficient

1 (L7)

(b) (i) *embryo*: stage 3

accept '3'

foetus: stage 4

accept '4'

***both** answers are required for the mark*

1 (L7)

(ii) uterus

accept 'womb'

1 (L6)

(c) (i) any **two** from

- a reference to genes **or** genetic material **or** DNA **or** chromosomes

  accept for one mark 'to be the same as someone else you need exactly the same genes'

- it has a different combination (of genes)

- it is made from an egg and sperm

  *accept for two marks 'it only has half the DNA **or** chromosomes from each parent'*

- (it gets genetic information) from both parents

2 (L7)

(ii)

human characteristic	inherited only	inherited and affected by environmental conditions
eye colour	✓	
skin colour		✓
weight		✓

***all three** ticks are required for the mark*
if there is more than one tick in any row, award no mark

1 (L7)

[7]